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Method of making amphi-FET structure and method of designing

Abstract

A semiconductor device includes a substrate. The semiconductor device further includes a first gate structure on a first side of the substrate. The semiconductor device further includes a second gate structure on a second side of the substrate, wherein the first side is opposite the second side. The semiconductor device further includes a gate via extending through the substrate, wherein the gate via directly connects to the first gate structure, and the gate via directly connects to the second gate structure.

Inventors: Lai; Chih-Yu (Hsinchu, TW), Chen; Chih-Liang (Hsinchu, TW), Lu; Chi-Yu (Hsinchu, TW), Ciou; Shang-Syuan (Hsinchu, TW), Zhuang; Hui-Zhong (Hsinchu, TW), Tsai; Ching-Wei (Hsinchu, TW), Chang; Shang-Wen (Hsinchu, TW)

Applicant: TAIWAN SEMICONDUCTOR MANUFACTURING COMPANY, LTD.
(Hsinchu, TW)

Family ID: 82422755

Assignee: TAIWAN SEMICONDUCTOR MANUFACTURING COMPANY, LTD.
(Hsinchu, TW)

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Primary Examiner: Han; Jonathan

Attorney, Agent or Firm: Hauptman Ham, LLP

Background/Summary

RELATED APPLICATIONS (1) This application is a continuation of U.S. application Ser. No. 18/360,539, filed Jul. 27, 2023, now U.S. Pat. No. 12,009,362, issued Jun. 11, 2024, which is a divisional of U.S. application Ser. No. 17/214,194, filed Mar. 26, 2021, now U.S. Pat. No. 11,764,213, issued Sep. 19, 2023, which are incorporated herein by reference in their entireties.

BACKGROUND

(1) As technology nodes continue to shrink routing for interconnect structures becomes more difficult. Three dimensional integrated circuits (3DICs) involve stacking devices in a vertical direction and electrically connecting the devices together, for example, using a through silicon via (TSV). In a 3DIC structure, a device is formed on one side of a substrate. The 3DIC reduces an area for the IC in a planar direction.

(2) In another approach, interconnect structures, such as routing lines and power lines are formed on one side of a substrate and connected to a device on an opposite side of the substrate, for example using a TSV. The interconnect structures on an opposite side of the substrate from the device increases an area usable for routing, which increases routing options and reduces an area of the IC in the planar direction.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) Aspects of the present disclosure are best understood from the following detailed description when read with the accompanying figures. It is noted that, in accordance with the standard practice in the industry, various features are not drawn to scale. In fact, the dimensions of the various features may be arbitrarily increased or reduced for clarity of discussion.

(2) FIG. 1 is a cross-sectional view of an amphi-FET, in accordance with some embodiments.

(3) FIG. 2 is a cross-sectional view of an amphi-FET, in accordance with some embodiments.

(4) FIG. 3 is a flowchart of a method of making an amphi-FET, in accordance with some embodiments.

(5) FIG. 4 is a series of cross-sectional views of an amphi-FET structure at various stages of manufacture, in accordance with some embodiments.

(6) FIG. 5A is a top view of an amphi-FET structure, in accordance with some embodiments.

(7) FIG. 5B is a cross-sectional view of the amphi-FET structure of FIG. 5 taken along the line B-B', in accordance with some embodiments.

(8) FIG. 5C is a cross-sectional view of the amphi-FET structure of FIG. 5 taken along the line C-C', in accordance with some embodiments.

(9) FIG. 6A is a schematic flowchart of a method of converting a layout of a FET structure into a layout of an amphi-FET structure, in accordance with some embodiments.

(10) FIG. 6B is perspective views of a FET structure converted into an amphi-FET structure, in accordance with some embodiments.

(11) FIG. 7A is a schematic flowchart of a method of converting a layout of a FET structure into a layout of an amphi-FET structure, in accordance with some embodiments.

(12) FIG. 7B is perspective views of a FET structure converted into an amphi-FET structure, in accordance with some embodiments.

(13) FIGS. 7C-7E are perspective views of portions of the amphi-FET structure in FIG. 7B, in accordance with some embodiments.

(14) FIG. 8 is view of converting neighboring cells in a layout of a FET structure to a layout of an amphi-FET structure, in accordance with some embodiments.

(15) FIG. 9 is a view of a layout of an amphi-FET structure, in accordance with some embodiments.

(16) FIG. 10 is a view of a layout of an amphi-FET structure, in accordance with some embodiments.

(17) FIG. 11 is a view of a layout of an amphi-FET structure, in accordance with some embodiments.

(18) FIG. 12 is a view of a layout of an amphi-FET structure, in accordance with some embodiments.

(19) FIG. 13 is a view of a layout of an amphi-FET structure, in accordance with some embodiments.

(20) FIG. 14 is a block diagram of an integrated circuit (IC) manufacturing system, and an IC manufacturing flow associated therewith, in accordance with some embodiments.

(21) FIG. 15 is a block diagram of an electronic design automation (EDA) system, in accordance with some embodiments.

DETAILED DESCRIPTION

(22) The following disclosure provides many different embodiments, or examples, for implementing different features of the provided subject matter. Specific examples of components, values, operations, materials, arrangements, or the like, are described below to simplify the present disclosure. These are, of course, merely examples and are not intended to be limiting. Other components, values, operations, materials, arrangements, or the like, are contemplated. For example, the formation of a first feature over or on a second feature in the description that follows may include embodiments in which the first and second features are formed in direct contact, and may also include embodiments in which additional features may be formed between the first and second features, such that the first and second features may not be in direct contact. In addition, the present disclosure may repeat reference numerals and/or letters in the various examples. This repetition is for the purpose of simplicity and clarity and does not in itself dictate a relationship between the various embodiments and/or configurations discussed.

(23) Further, spatially relative terms, such as “beneath,” “below,” “lower,” “above,” “upper” and the like, may be used herein for ease of description to describe one element or feature's relationship to another element(s) or feature(s) as illustrated in the figures. The spatially relative terms are intended to encompass different orientations of the device in use or operation in addition to the orientation depicted in the figures. The apparatus may be otherwise oriented (rotated 90 degrees or at other orientations) and the spatially relative descriptors used herein may likewise be interpreted accordingly.

(24) As technology nodes continue to shrink, three dimensional integrated circuits (3DICs), fin field effect transistors (FinFETs), gate all around (GAA) transistors, and backside routing structures alone are unable to keep up with the demand for reduced device area. Amphi-field effect transistors (amphi-FETs) are usable to improve routing options for interconnect structures, which permits further device size reduction. An amphi-FET includes active devices formed on both sides of a substrate. In contrast, a 3DIC includes two devices formed on separate substrates; and the separate substrates are then bonded together. The improved routing options in an amphi-FET means that increasing device size unnecessarily to accommodate connections between elements in the device is reduced or avoided.

(25) The active devices in the amphi-FET are electrically connected using conductive elements extending through the substrate. For example, a gate structure on a first side of the substrate is electrically connected to a gate structure on a second side of the substrate by a gate via extending through the substrate. Similarly, a source/drain (S/D) contact on the first side of the substrate is electrically connected to an S/D contact on the second side of the substrate by an S/D connect via extending through the substrate. Due to these connections through the substrate, a signal, such as a power signal applied on one side of the substrate is able to transfer to the other side of the substrate

without including additional routing or through substrate via (TSV) structures. For example, a ground signal connected to an S/D contact on the first side of the substrate is transferred to the second side of the substrate by the S/D connect via to the S/D contact on the second side of the substrate. The ground voltage is then usable on the second side of the substrate without providing a power rail for the ground voltage (or a TSV structure) on the second side of the substrate.

(26) Integrated circuits (ICs) are often designed using cells, which include active devices and connection structures in order to implement an intended function. In some embodiments, a cell includes a single active device. In some embodiments, a cell includes multiple active devices. The ability to efficiently share signals and power from one side of the substrate to the other side of the substrate makes more routing tracks in a cell available for routing signals within the cell and into and out of the cell. Thus, a size of the cell is not unnecessarily increased merely to provide sufficient connections for the cell to function properly. In some instances, a gate density of a device including amphi-FETs is greater than 1.5 times a gate density of a device including active devices on a single side of the substrate. Gate density is a measure of how closely gate structures in a cell are spaced from one another. Gate density is commonly used to describe how efficiently the space within the cell is utilized. Increasing the cell size merely to provide routing options reduces gate density. In some instances, the gate density increase due to the use of amphi-FETs is more than 1.6 times the gate density of a device including active devices on a single side of the substrate.

(27) This significant size reduction helps to keep pace with Moore's Law and also helps with reduced power consumption and increased device speed. The reduced power consumption is a result of having shorter distances between elements, so less power is lost to resistance and heating of conductive lines and vias in the amphi-FET device. The increased device speed results from being able to increase a size of active regions of the cell because less space within the cell is occupied by interconnect elements.

(28) FIG. 1 is a cross-sectional view of an amphi-FET **100** in accordance with some embodiments. The amphi-FET **100** includes a substrate **110**. A first cell **120** is on a first side of the substrate **110**. A second cell **130** is on a second side of the substrate **110** opposite to the first side. The first cell **120** is electrically connected to the second cell **130** through the substrate **110**, as discussed below. In some embodiments, the first cell **120** has a same functionality as the second cell **130**. In some embodiments, the first cell **120** has a different functionality from the second cell **130**. In some embodiments, the first cell **120** and the second cell **130** combine to implement a designed functionality. One of ordinary skill in the art will recognize that two cells are included in FIG. 1 as an example, and that the current description is not limited to merely two cells.

(29) FIG. 2 is a cross-sectional view of an amphi-FET **200** in accordance with some embodiments. In some embodiments, the amphi-FET **200** is a more detailed view of the amphi-FET **100** (FIG. 1). The amphi-FET **200** includes a substrate **210**. A first cell **220** is on a first side of the substrate **210** and a second cell **230** is on a second side of the substrate **210** opposite to the first side. The first cell **220** is connected to the second cell **230** by an S/D connect via **212** extending through the substrate **210**, and by a gate via **214** extending through the substrate **210**. The S/D connect via **212** has parallel sidewalls and a substantially uniform width. The gate via **214** has a tapered profile. In some embodiments, the S/D connect via **212** has a tapered profile. In some embodiments, the gate via **214** has parallel sidewalls and a substantially uniform width. One of ordinary skill in the art would recognize that any combination of shapes for the S/D connect via **212** and the gate via **214** is within the scope of this description.

(30) The first cell **220** includes a first active region **242**. In some embodiments, the first active region **242** includes nanosheets (NS) for forming a gate all around (GAA) transistor. A first gate contact **250** is electrically connected to a gate structure associated with the first active region **242**. A first via **252** electrically connects the first gate contact **250** to a first conductive line **254** of a first side interconnect structure. In some embodiments, the first conductive line **254** is usable to carry a signal or power to the gate structure in the first active region **242** through the first via **252** and the

first gate contact **250**.

(31) The first cell **220** further includes a first S/D contact **260** electrically connected to an S/D electrode associated with the first active region **242**. A second via **262** electrically connects the first S/D contact **260** to a second conductive line **264** of the first side interconnect structure. In some embodiments, the second conductive line **264** is usable to carry a signal or power to the S/D electrode in the first active region **242** through the second via **262** and the first S/D contact **260**.

(32) The second cell **230** includes a second active region **244**. In some embodiments, the second active region **244** includes NS for forming a GAA transistor. A second gate contact **270** is electrically connected to a gate structure associated with the second active region **244**. A third via **272** electrically connects the second gate contact **270** to a third conductive line **274** of a second side interconnect structure. In some embodiments, the third conductive line **274** is usable to carry a signal or power to the gate structure in the second active region **244** through the third via **272** and the second gate contact **270**.

(33) The second cell **230** further includes a second S/D contact **280** electrically connected to an S/D electrode associated with the second active region **244**. A fourth via **282** electrically connects the second S/D contact **280** to a fourth conductive line **284** of the second side interconnect structure. In some embodiments, the fourth conductive line **284** is usable to carry a signal or power to the S/D electrode in the second active region **244** through the fourth via **282** and the second S/D contact **280**.

(34) The third conductive line **274** is electrically connected to the first conductive line **254** through the third via **272**, the second gate contact **270**, the gate structure in the second active region **244**, the gate via **214**, the gate structure in the first active region **242**, the first gate contact **250** and the first via **252**. This connection path is more direct than routing a connection path around the first active region **242** and the second active region **244**. As a result, resistance of the amphi-FET **200** is reduced and a size of each of the first cell **220** and the second cell **230** is reduced in comparison with other structures that do not include the gate via **214**.

(35) The fourth conductive line **284** is electrically connected to the second conductive line **264** through the fourth via **282**, the second S/D contact **280**, the S/D electrode in the second active region **244**, the S/D connect via **212**, the S/D electrode in the first active region **242**, the first S/D contact **260** and the second via **262**. This connection path is more direct than routing a connection path around the first active region **242** and the second active region **244**. As a result, resistance of the amphi-FET **200** is reduced and a size of each of the first cell **220** and the second cell **230** is reduced in comparison with other structures that do not include the S/D connect via **212**.

(36) In some embodiments, substrate **210** includes an elementary semiconductor including silicon or germanium in crystal, polycrystalline, or an amorphous structure; a compound semiconductor including silicon carbide, gallium arsenic, gallium phosphide, indium phosphide, indium arsenide, and indium antimonide; an alloy semiconductor including SiGe, GaAsP, AlInAs, AlGaAs, GaInAs, GaInP, and GaInAsP; any other suitable material; or combinations thereof. In some embodiments, the alloy semiconductor substrate has a gradient SiGe feature in which the Si and Ge composition changes from one ratio at one location to another ratio at another location of the gradient SiGe feature. In some embodiments, the alloy SiGe is formed over a silicon substrate. In some embodiments, substrate **210** is a strained SiGe substrate. In some embodiments, the semiconductor substrate has a semiconductor on insulator structure, such as a silicon on insulator (SOI) structure. In some embodiments, the semiconductor substrate includes a doped epi layer or a buried layer. In some embodiments, the compound semiconductor substrate has a multilayer structure or the substrate includes a multilayer compound semiconductor structure.

(37) In some embodiments, a number of NS in the first active region **242** is equal to a number of NS in the second active region **244**. In some embodiments, the number of NS in the first active region **242** is different from the number of NS in the second active region **244**. In some embodiments, at least one of the first active region **242** or the second active region **244** includes a

transistor structure different from a GAA transistor, such as a FinFET or metal-oxide-semiconductor FET (MOSFET). While the cross-sectional view of FIG. 2 includes the S/D connect via **212** and the gate via **214** extending through the first active region **242** and the second active region **244**, the gate via **214** is not electrically shorted to the S/D connect via **212** by the first active region **242** or the second active region **244**. One of ordinary skill in the art would recognize that the gate via **214** and the S/D connect via **212** electrically connect to portions of the first active region **242** and the second active region **244** as made clear by the description below.

(38) In some embodiments, the S/D connect via **212** and the gate via **214** independently include a fill material including cobalt (Co), tungsten (W), copper (Cu), ruthenium (Ru), titanium (Ti), tantalum (Ta), or alloys thereof, or other conductive materials. In some embodiments, the S/D connect via **212** and the gate via **214** include a liner surrounding the fill material. In some embodiments, the liner includes titanium nitride (TiN), titanium (Ti), ruthenium (Ru), cobalt (Co), tantalum (Ta), tantalum nitride (Ta₂N₃), or alloys thereof, or other liner materials. In some embodiments, a composition of the S/D connect via **212** is a same composition as the gate via **214**. In some embodiments, the composition of the S/D connect via **212** is different from the composition of the gate via **214**. In some embodiments, the S/D connect via **212** connects to the S/D electrode of the first active region **242** and the second active region **244** through a silicide layer (not shown). In some embodiments, the gate via **214** connects to the gate structure of the first active region **242** and the second active region **244** through a silicide layer (not shown).

(39) In some embodiments, the first gate contact **250** and the second gate contact **270** independently include a fill material including cobalt (Co), tungsten (W), copper (Cu), ruthenium (Ru), titanium (Ti), tantalum (Ta), or alloys thereof, or other conductive materials. In some embodiments, the first gate contact **250** and the second gate contact **270** include a liner surrounding the fill material. In some embodiments, the liner includes titanium nitride (TiN), titanium (Ti), ruthenium (Ru), cobalt (Co), tantalum (Ta), tantalum nitride (Ta₂N₃), or alloys thereof, or other liner materials. In some embodiments, a composition of the first gate contact **250** is a same composition as the second gate contact **270**. In some embodiments, the composition of the first gate contact **250** is different from the composition of the second gate contact **270**. In some embodiments, each of the first gate contact **250** and the second gate contact **270** have a same composition as both the S/D connect via **212** and the gate via **214**. In some embodiments, at least one of the first gate contact **250** or the second gate contact **270** has a different composition from at least one of the S/D connect via **212** or the gate via **214**. In some embodiments, the first gate contact **250** and the second gate contact **270** independently connect to the gate structure of the corresponding first active region **242** and the second active region **244**, through a silicide layer (not shown).

(40) In some embodiments, the first S/D contact **260** and the second S/D contact **280** independently include a fill material including cobalt (Co), tungsten (W), copper (Cu), ruthenium (Ru), titanium (Ti), tantalum (Ta), or alloys thereof, or other conductive materials. In some embodiments, the first S/D contact **260** and the second S/D contact **280** include a liner surrounding the fill material. In some embodiments, the liner includes titanium nitride (TiN), titanium (Ti), ruthenium (Ru), cobalt (Co), tantalum (Ta), tantalum nitride (Ta₂N₃), or alloys thereof, or other liner materials. In some embodiments, a composition of the first S/D contact **260** is a same composition as the second S/D contact **280**. In some embodiments, the composition of the first S/D contact **260** is different from the composition of the second S/D contact **280**. In some embodiments, each of the first S/D contact **260** and the second S/D contact **280** have a same composition as the S/D connect via **212**, the gate via **214**, the first gate contact **250** and the second gate contact **270**. In some embodiments, at least one of the first S/D contact **260** or the second S/D contact **280** has a different composition from at least one of the S/D connect via **212**, the gate via **214**, the first gate contact **250** or the second gate contact **270**. In some embodiments, the first S/D contact **260** and the second S/D contact **280** independently connect to the S/D electrode of the first active region **242** and the second active region **244**, respectively, through a silicide layer (not shown).

(41) In some embodiments, the first via **252**, the second via **262**, the third via **272** and the fourth via **282** independently include a fill material including cobalt (Co), tungsten (W), copper (Cu), ruthenium (Ru), titanium (Ti), tantalum (Ta), or alloys thereof, or other conductive materials. In some embodiments, the first via **252**, the second via **262**, the third via **272** and the fourth via **282** include a liner surrounding the fill material. In some embodiments, the liner includes titanium nitride (TiN), titanium (Ti), ruthenium (Ru), cobalt (Co), tantalum (Ta), tantalum nitride (Ta₂N₃), or alloys thereof, or other liner materials. In some embodiments, a composition of the first via **252** is a same composition as each of the second via **262**, the third via **272** and the fourth via **282**. In some embodiments, the composition of the first via **252** is different from the composition of at least one of the second via **262**, the third via **272** or the fourth via **282**. In some embodiments, each of the first via **252**, the second via **262**, the third via **272** and the fourth via **282** have a same composition as the S/D connect via **212**, the gate via **214**, the first gate contact **250**, the second gate contact **270**, the first S/D contact **260** and the second S/D contact **280**. In some embodiments, at least one of the first via **252**, the second via **262**, the third via **272** or the fourth via **282** has a different composition from at least one of the S/D connect via **212**, the gate via **214**, the first gate contact **250**, the second gate contact **270**, the first S/D contact **260** or the second S/D contact **280**.

(42) In some embodiments, the first conductive line **254**, the second conductive line **264**, the third conductive line **274** and the fourth conductive line **284** independently include a fill material including cobalt (Co), tungsten (W), copper (Cu), ruthenium (Ru), titanium (Ti), tantalum (Ta), or alloys thereof, or other conductive materials. In some embodiments, the first conductive line **254**, the second conductive line **264**, the third conductive line **274** and the fourth conductive line **284** include a liner surrounding the fill material. In some embodiments, the liner includes titanium nitride (TiN), titanium (Ti), ruthenium (Ru), cobalt (Co), tantalum (Ta), tantalum nitride (Ta₂N₃), or alloys thereof, or other liner material. In some embodiments, a composition of the first conductive line **254** is a same composition as each of the second conductive line **264**, the third conductive line **274** and the fourth conductive line **284**. In some embodiments, the composition of the first conductive line **254** is different from the composition of at least one of the second conductive line **264**, the third conductive line **274** or the fourth conductive line **284**. In some embodiments, each of the first conductive line **254**, the second conductive line **264**, the third conductive line **274** and the fourth conductive line **284** have a same composition as the S/D connect via **212**, the gate via **214**, the first gate contact **250**, the second gate contact **270**, the first S/D contact **260**, the second S/D contact **280**, the first via **252**, the second via **262**, the third via **272** and the fourth via **282**. In some embodiments, at least one of the first conductive line **254**, the second conductive line **264**, the third conductive line **274** or the fourth conductive line **284** has a different composition from at least one of the S/D connect via **212**, the gate via **214**, the first gate contact **250**, the second gate contact **270**, the first S/D contact **260**, the second S/D contact **280**, the first via **252**, the second via **262**, the third via **272** or the fourth via **282**.

(43) FIG. 3 is a flowchart of a method **300** of making an amphi-FET in accordance with some embodiments. The operations of the method **300** are able to be performed in a variety of sequences in order to produce the amphi-FET. Some possible variations will be discussed below.

(44) Method **300** includes operation **305**, in which a first cell is formed on a first side of substrate. The first cell is formed using a series of deposition and patterning processes. In some embodiments, the deposition processes include chemical vapor deposition (CVD), physical vapor deposition (PVD), atomic layer deposition (ALD), sputtering or another suitable deposition process. In some embodiments, the patterning processes include lithographic and etching processes, such as dry or wet etching. In some embodiments, the first cell includes NS and an interconnect structure, similar to the first cell **220** (FIG. 2).

(45) In operation **310**, an S/D connect via is formed through the substrate. The S/D connect via is formed by depositing a sacrificial layer, such as a dielectric layer over at least one side of the substrate. In some embodiments where the S/D connect via is formed after the first cell, the

sacrificial layer is formed on a single side of the substrate. In some embodiments where the first cell is formed after the S/D connect via, the sacrificial layer is formed on both sides of the substrate. The sacrificial layer and the substrate are etched to form an opening extending through the substrate. In embodiments that include forming the first cell prior to forming the S/D connect via, the opening exposes a portion of an S/D electrode of the first cell. The opening is then filled by one or more deposition processes in order to form the S/D connect. In some embodiments, the deposition processes include CVD, PVD, ALD, sputtering or another suitable deposition process. In some embodiments, the S/D connect via is similar to the S/D connect via **212** (FIG. 2). In some embodiments, a removal process, such as etching or planarization is performed to remove material of the S/D connect via over a surface of the sacrificial layer farthest from the substrate. In some embodiments, the sacrificial layer is removed from at least one side of the substrate following formation of the S/D connect via.

(46) In operation **315**, a gate via is formed through the substrate. In some embodiments, the gate via is formed simultaneously with the formation of the S/D connect via. The gate via is formed by depositing a sacrificial layer, such as a dielectric layer over at least one side of the substrate. In some embodiments where the gate via is formed after the first cell, the sacrificial layer is formed on a single side of the substrate. In some embodiments where the first cell is formed after the gate via, the sacrificial layer is formed on both sides of the substrate. The sacrificial layer and the substrate are etched to form an opening extending through the substrate. In embodiments that include forming the first cell prior to forming the gate via, the opening exposes a portion of a gate structure of the first cell. The opening is then filled by one or more deposition processes in order to form the S/D connect. In some embodiments, the deposition processes include CVD, PVD, ALD, sputtering or another suitable deposition process. In some embodiments, the gate via is similar to the gate via **214** (FIG. 2). In some embodiments, a removal process, such as etching or planarization is performed to remove material of the gate via over a surface of the sacrificial layer farthest from the substrate. In some embodiments, the sacrificial layer is removed from at least one side of the substrate following formation of the gate via.

(47) In operation **320**, the substrate is flipped. In some embodiments, the substrate is flipped using a robot arm and/or a vacuum chuck. Flipping the substrate exposes a second side of the substrate for processes. In some embodiments where the S/D connect via and/or the gate via is formed after the first cell, the substrate is flipped prior to forming the S/D connect via and/or the gate via. In some embodiments wherein the S/D connect via and/or the gate via is formed prior to the first cell, the substrate is flipped after forming the first cell.

(48) In operation **325**, a second cell is formed on the second side of the substrate. The second cell is formed using a series of deposition and patterning processes. In some embodiments, the deposition processes include CVD, PVD, ALD, sputtering or another suitable deposition process. In some embodiments, the patterning processes include lithographic and etching processes, such as dry or wet etching. In some embodiments, the second cell includes NS and an interconnect structure, similar to the second cell **230** (FIG. 2). Forming the second cell includes forming an S/D electrode of the second cell electrically connected to the S/D connect via and forming a gate structure of the second cell electrically connected to the gate via. In some embodiments, a functionality of the first cell is a same functionality as the second cell. In some embodiments, the first cell has different functionality from the second cell. In some embodiments, the first cell and the second cell are used in combination to implement a designed functionality.

(49) FIG. 4 is a series of cross-sectional views of an amphi-FET structure **400** at various stages of manufacture in accordance with some embodiments. FIG. 4 includes a first production track **402** and a second production track **404**. Both the first production track **402** and the second production track **404** product a same final structure as indicated in FIG. 4.

(50) In the first production track **402**, the first cell **220** is formed on a first side of the substrate **210**. The substrate **210** is then flipped and the S/D connect via **212** and the gate via **214** are formed to

electrically connect to the first cell **220** through the substrate **210**. The second cell **230** is then formed on the second side of the substrate **210** to electrically connect to the first cell **220** through the S/D connect via **212** and the gate via **214**. In some embodiments, the first production track **402** is implemented by performing operations **305**, **320**, **310**, **315** and **325** of method **300** (FIG. 3) in this order. In some embodiments, operations **310** and **315** are performed simultaneously.

(51) In the second production track **404**, the S/D connect via **212** and the gate via **214** are formed through the substrate **210**. The first cell **220** is then formed on the first side of the substrate **210** to electrically connect to the S/D connect via **212** and the gate via **214**. The substrate **210** is then flipped and the second cell **230** is formed on the second side of the substrate **210** to electrically connect to the first cell **220** through the S/D connect via **212** and the gate via **214**. In some embodiments, the second production track **404** is implemented by performing operations **310**, **315**, **305**, **320** and **325** of method **300** (FIG. 3) in this order. In some embodiments, operations **310** and **315** are performed simultaneously.

(52) FIG. 5A is a top view of an amphi-FET **500** in accordance with some embodiments. The amphi-FET **500** includes a first active region **515**. In some embodiments, the first active region **515** includes one or more NS. A plurality of first S/D electrodes **520** extend across the first active region **515**. A plurality of first gate structures **530** also extend across the first active region **515** in an alternating fashion with the plurality of first S/D electrodes **520**. The first S/D electrode **520** between the two first gate structures **530** is a shared first S/D electrode **520** for both of the first gate structures **530**. FIG. 5A includes a view of the portions of the amphi-FET **500** only on a first side of a substrate. The substrate and portions of the amphi-FET **500** on the second side of the substrate are obscured by the structures in FIG. 5A. Conductive lines and vias are included in FIGS. 5B and 5C, but are not included in FIG. 5A for the sake of clarity.

(53) FIG. 5B is a cross-sectional view of the amphi-FET **500** taken along the line B-B' in accordance with some embodiments. The amphi-FET **500** includes a substrate **510** and the first active area **515** is on a first side of the substrate **510**. The first active area **515** includes one or more NS. The first S/D electrode **520** surrounds the portion of the first active region **515** seen in the cross-sectional view and extends between the first active region **515** and the substrate **510**. A second active region **515'** is on a second side of the substrate **510**. The second active region **515'** includes one or more NS. A second S/D electrode **520'** surrounds the portion of the second active region **515'** seen in the cross-sectional view and extends between the second active region **515'** and the substrate **510**. The first S/D electrode **520** is electrically connected to the second S/D electrode **520'** by an S/D connect via **512** that extends through the substrate **510**.

(54) A first via **540** electrically connects the first S/D electrode **520** to a first power rail **550**. In some embodiments, the first power rail **550** carries a reference voltage, e.g., ground. In some embodiments, the first power rail **550** carries a source voltage, e.g., VDD. In some embodiments, the first via **540** is omitted and the first power rail **550** is not directly connected to the first S/D electrode **520**. A second via **545** electrically connects the first S/D electrode **520** to a second conductive line **555b**. The second conductive line **555b** is able to carry a signal to or from the first S/D electrode **520**. In some embodiments, the second via **545** is omitted and the second conductive line **555b** is not directly connected to the first S/D electrode **520**. A first conductive line **555a** is between the first power rail **550** and the second conductive line **555b**. In some embodiments, the first conductive line **555a** is connected to the first S/D electrode **520** by a via to carry a signal to or from the first S/D electrode **520**. In some embodiments, the first S/D electrode **520** is electrically connected to the first power rail **550** and the second S/D electrode **520'** is electrically connected to the second power rail **550'**.

(55) A third via **545'** electrically connects the second S/D electrode **520'** to a third conductive line **555a'**. The third conductive line **555a'** is able to carry a signal to or from the second S/D electrode **520'**. In some embodiments, the third via **545'** is omitted and the third conductive line **555a'** is not directly connected to the second S/D electrode **520'**. The third conductive line **555a'** is between a

second power rail **550'** and a fourth conductive line **555b'**. In some embodiments, the fourth conductive line **555b'** is connected to the second S/D electrode **520'** by a via to carry a signal to or from the second S/D electrode **520'**.

(56) In some embodiments, the second power rail **550'** carries a reference voltage, e.g., ground. In some embodiments, the second power rail **550'** carries a source voltage, e.g., VDD. In some embodiments, the second power rail **550'** is connected to the second S/D electrode **520'** by a via.

(57) The material and formation method of the first via **540**, the second via **545**, the third via **545'**, the first power rail **550**, the first conductive line **555a**, the second conductive line **555b**, the second power rail **550'**, the third conductive line **555a'** and the fourth conductive line **555b'** are similar to materials and formation methods for similar elements described above with respect to amphi-FET **200** (FIG. 2). The material and formation method of the S/D connect via **512** is similar to S/D connect via **212** (FIG. 2).

(58) The first S/D electrode **520** and the second S/D electrode **520'** conductive material surround S/D regions of the corresponding active regions. In some embodiments, the S/D regions include doped semiconductor materials. In some embodiments, the S/D regions include silicon. In some embodiments, the S/D regions include a strained material, such as silicon-germanium (SiGe). In some embodiments, the S/D regions are formed by epitaxial processes. In some embodiments, the S/D regions are formed by ion implantation. In some embodiments, the first S/D electrode **520** and the second S/D electrode **520'** are formed by deposition. In some embodiments, the first S/D electrode **520** and the second S/D electrode **520'** independently include cobalt (Co), tungsten (W), copper (Cu), ruthenium (Ru), titanium (Ti), tantalum (Ta), or alloys thereof, or other conductive materials. In some embodiments, a composition of the first S/D electrode **520** is a same composition as the second S/D electrode **520'**. In some embodiments, the composition of the first S/D electrode **520** is different from the composition of the second S/D electrode **520'**. In some embodiments, the first S/D electrode **520** and the second S/D electrode **520'** include a silicide layer.

(59) A width of the S/D connect via **512** in the substrate **510** is equal to a width of the second S/D electrode **520'**. The width is measured in a direction parallel to a top surface of the substrate **510**. Maximizing a size of the S/D connect via **512** helps to minimize resistance in the S/D connect via **512**, which improves power consumption. The width of the second S/D electrode **520'** is less than a width of the first S/D electrode **520**. In some embodiments, the width of the second S/D electrode **520'** is equal to the width of the first S/D electrode **520**.

(60) A first dimension **D1** of the active region **515** in a first direction perpendicular to the top surface of the substrate **510** ranges from about 0.8 times to about 15 times a minimum gate width. If the first dimension **D1** is too small, then resistance within a channel of the amphi-FET **500** increases and impacts device performance, in some instances. If the first dimension **D1** is too large, then a size of the amphi-FET **500** is increased without significant improvement in device performance, in some instances. The minimum gate width is also referred to as a critical dimension (CD), in some instances. The minimum gate width is a smallest size that is reliably produced during a manufacturing process. One of ordinary skill in the art would recognize that different technology nodes have different manufacturing processes and are able to produce different minimum gate widths.

(61) A ratio of the first dimension **D1** and a second dimension **D2** of the first power rail **550** in a second direction parallel to the top surface of the substrate **510** ranges from about 1 to about 5. If the second dimension **D2** is too small, then resistance in the first power rail **550** increases to a level that negatively impacts power consumption and uniform power distribution within the device, in some instances. If the second dimension **D2** is too large, then a size of the amphi-FET **500** is increased without a significant improvement in performance, in some instances. In some embodiments, the second power rail **550'** has a same dimension as the first power rail **550**.

(62) A ratio of the first dimension **D1** and a third dimension **D3** of the second conductive line **555b** in the second direction ranges from about 0.5 to about 3. If the third dimension **D3** is too small,

then resistance in the second conductive line **555b** increases to a level that negatively impacts power consumption and signal reliability within the device, in some instances. If the third dimension **D3** is too large, then a size of the amphi-FET **500** is increased without a significant improvement in performance, in some instances. In some embodiments, at least one of the first conductive line **555a**, the third conductive line **555a'** or the fourth conductive line **555b'** has a same dimension as the second conductive line **555b**.

(63) A ratio of the first dimension **D1** and a fourth dimension **D4** of the first via **540** in the first direction ranges from about 2 to about 6. If the fourth dimension **D4** is too small, then spacing between the first power rail **550** and the first S/D electrode **520** increases a risk of short circuiting within the device, in some instances. If the fourth dimension **D4** is too large, then a size of the amphi-FET **500** is increased without a significant improvement in performance, in some instances. In some embodiments, at least one of the second via **545** or the third via **545'** has a same dimension as the first via **540**. In some embodiments, a sixth dimension **D6** is equal to the fourth dimension **D4**. In some embodiments, the sixth dimension **D6** is different from the fourth dimension **D4**.

(64) A ratio of the first dimension **D1** and a fifth dimension **D5** from a surface of the first S/D electrode **520** farthest from the substrate **510** to a surface of the second S/D electrode **520'** farthest from the substrate in the first direction ranges from about 10 to about 60. If the fifth dimension **D5** is too small, then reliability of manufacturing the amphi-FET **500** is negatively impacted, in some instances. If the fifth dimension **D5** is too large, then a size of the amphi-FET **500** is increased without a significant improvement in performance, in some instances.

(65) A ratio of the first dimension **D1** and a seventh dimension **D7** of a space between the third conductive line **555a'** and the fourth conductive line **555b'** in the second direction ranges from about 0.5 to about 3. If the seventh dimension **D7** is too small, then a risk of short circuit within the device or negative impacts from parasitic capacitance impact device performance, in some instances. If the seventh dimension **D7** is too large, then a size of the amphi-FET **500** is increased without a significant improvement in performance, in some instances. In some embodiments, a spacing between other combinations of adjacent conductive lines and/or power rails has a same dimension as the seventh dimension **D7**.

(66) FIG. 5C is a cross-sectional view of the amphi-FET **500** taken along line C-C' in accordance with some embodiments. Same elements from FIG. 5B have a same reference number and the description of these elements is omitted for the sake of brevity. A first gate structure **530** surrounds the portion of the first active region **515** visible in the cross-sectional view and extends between the first active region **515** and the substrate **510**. A second gate structure **530'** surrounds the portion of the second active region **515'** visible in the cross-sectional view and extends between the second active region **515'** and the substrate **510**. The first gate structure **530** is electrically connected to the second gate structure **530'** by gate via **514**. In some embodiments, the gate via **514** is similar to the gate via **214** (FIG. 2). A first via **560** electrically connects the second conductive line **555b** to the first gate structure **530** in order to carry a signal into or out of the first gate structure **530**. In some embodiments, the first via **560** is omitted and the first gate structure **530** is not directly connected to the second conductive line **555b**. A second via **560'** electrically connects the third conductive line **555a'** to the second gate structure **530'** to carry a signal into or out of the second gate structure **530'**. In some embodiments, the second via **560'** is omitted and the second gate structure **530'** is not directly connected to the third conductive line **555a'**. While FIG. 5C includes a connection to both the first gate structure **530** and the second gate structure **530'**, one of ordinary skill in the art would recognize that in most situations the gate structures **530** and **530'** will be driven from a single side. That is, in most situations only one of the first via **560** or the second via **560'** will be present. In some embodiments where the amphi-FET **500** is in a constant ON or OFF state, the first gate structure **530** is connected to the first power rail **550** by a via. In some embodiments where the amphi-FET **500** is in a constant ON or OFF state, the second gate structure **530'** is connected to the second power rail **550'** by a via.

(67) In some embodiments, a size of the gate via **514** is a same size as the first via **560**. Having the gate via **514** have a same size as the first via **560** reduces production cost by minimizing a number of masks used to produce the amphi-FET **500**. Having the gate via **514** have the same size as the first via **560** also helps to reduce manufacturing error because the openings for forming the first via **560** and the gate via **514** are self-aligned.

(68) The first gate structure **530** and the second gate structure **530'** independently include a gate dielectric layer and a gate electrode. The gate dielectric layer is between the corresponding active region and the corresponding gate electrode. In some embodiments, the gate dielectric layer includes silicon oxide and is formed by oxidizing an outer surface of the corresponding active region. In some embodiments, the gate dielectric layer includes silicon nitride or a high-k dielectric material. In some embodiments, the gate dielectric layer is formed by deposition, such as CVD, PVD, ALD or another suitable deposition process. The gate electrode includes a conductive material. In some embodiments, the gate electrode includes polysilicon. In some embodiments, the gate electrode comprises a metal, such as copper (Cu), cobalt (Co), aluminum (Al), tungsten (W), alloys or another suitable metal. In some embodiments, the gate electrode is formed by deposition, such as CVD, PVD, ALD or another suitable deposition process. In some embodiments, the first gate structure **530** and the second gate structure **530'** independently include an interfacial layer between the gate dielectric layer and the corresponding active region. In some embodiments, the first gate structure **530** and the second gate structure **530'** independently include a silicide layer.

(69) FIG. **6A** is a schematic flowchart of a method **600** of converting a layout of a FET structure into a layout of an amphi-FET structure in accordance with some embodiments. In some embodiments, the method **600** is executed by electronic design automation (EDA) system, such as EDA system **1500** (FIG. **15**).

(70) A FET layout **605** is received. The FET layout **605** is a layout for a structure designed to be formed on a single side of a substrate. In some embodiments, the FET layout **605** is for a GAA device, a FinFET device or a MOSFET device. In some embodiments, the FET layout **605** is received from a cell library. In some embodiments, the FET layout **605** is received from a user input. The FET layout **605** includes a plurality of routing tracks **606**. In some embodiments, at least one of the routing tracks **606** is a power rail. The FET layout **605** further includes a plurality of S/D electrodes including a first S/D electrode **608a**, a second S/D electrode **608b**, a third S/D electrode **608c** and a fourth S/D electrode **608d**. The FET layout **605** includes a first gate structure **609a** and a second gate structure **609b**.

(71) A vertical cut line **615** is added to the FET layout **605**. The vertical cut line **615** extends through the FET layout **605** in a direction perpendicular to the plurality of routing tracks **606**. The vertical cut line **615** is between the first S/D electrode **608a** and the third S/D electrode **608c**.

(72) The FET layout **605** is then divided into two cells. A first cell **620** includes the first S/D electrode **608a**, the second S/D electrode **608b** and the first gate structure **609a**. The location of routing tracks **606** in the first cell **620** is the same as the location of the routing tracks **606** in the FET layout **605**. A second cell **630** includes the third S/D electrode **608c**, the fourth S/D electrode **608d** and the second gate structure **609b**. The location of routing tracks **606** in the second cell **630** is the same as the location of the routing tracks **606** in the FET layout **605**. Maintaining the same location of the routing tracks **606** in the first cell **620** and the second cell **630** helps to simplify routing operations, such as automatic placement and routing (APR).

(73) The first cell **620** is then processed to be formed on a first side of a substrate **610** and the second cell **630** is processed to be formed on a second side of the substrate **610** opposite to the first side. In some embodiments, the substrate **610** is similar to the substrate **210** (FIG. **2**). Using the method **600**, an overall size of the amphi-FET device manufactured is reduced in comparison with manufacturing the FET layout **605** on a single side of the substrate.

(74) FIG. **6B** is perspective views of a FET structure converted into an amphi-FET structure in accordance with some embodiments. The substrate **610** is not included in FIG. **6B** for the sake of

clarity of the drawing. The left side of FIG. 6B is an example of the FET layout **605** for manufacturing on a single side of a substrate. The vertical cut line **615** is not part of the structure, but is included in FIG. 6B to indicate a location where the device is cut based on the method **600** (FIG. 6A). The right side of FIG. 6B is an example of an amphi-FET structure formed by the method **600**. As seen in FIG. 6B, elements in front of the vertical cut line **615** form the first cell **620** on a first side of a substrate. Elements behind the vertical cut line **615** form the second cell **630** on a second side of the substrate. A gate via, such as gate via **514** (FIG. 5C), would electrically connect the first gate structure **609a** to the second gate structure **609b** through the substrate. In some embodiments, an S/D connect via, such as S/D connect via **512** (FIG. 5B), electrically connects the first S/D electrode **608a** to the third electrode **608c** through the substrate. In some embodiments, the S/D connect via electrically connects the second S/D electrode **608b** to the fourth S/D electrode **608d** through the substrate. Since the vertical cut line **615** does not extend through any S/D electrodes of the FET layout **605**, in some embodiments, the amphi-FET does not include an S/D connect via.

(75) FIG. 7A is a schematic flowchart of a method **700** of converting a layout of a FET structure into a layout of an amphi-FET structure, in accordance with some embodiments. In some embodiments, the method **700** is executed by an electronic design automation (EDA) system, such as EDA system **1500** (FIG. 15).

(76) A FET layout **705** is received. The FET layout **705** is a layout for a structure designed to be formed on a single side of a substrate. In some embodiments, the FET layout **705** is for a GAA device, a FinFET device or a MOSFET device. In some embodiments, the FET layout **705** is received from a cell library. In some embodiments, the FET layout **705** is received from a user input. The FET layout **705** includes a plurality of routing tracks **706**. In some embodiments, at least one of the routing tracks **706** is a power rail. The FET layout **705** further includes a plurality of S/D electrodes includes a first S/D electrode **708a**, a second S/D electrode **708b** and a third S/D electrode **708c**. The FET layout **705** includes a gate structure **709**.

(77) A horizontal cut line **715** is added to the FET layout **705**. The horizontal cut line **715** extends through the FET layout **705** in a direction parallel to the plurality of routing tracks **706**. The horizontal cut line **715** is between the second S/D electrode **708b** and the third S/D electrode **708c**. The horizontal cut line **715** passes through the gate structure **709**.

(78) The FET layout **705** is then divided into two cells. A first cell **720** includes a first portion **708a'** of the first S/D electrode **708a**, the third S/D electrode **708c** and a first portion **709'** of the gate structure **709**. The location of routing tracks **706** in the first cell **720** are the same as the location of the routing tracks **706** in the FET layout **705**. A second cell **730** includes a second portion **708a''** of the first S/D electrode **708a**, the second S/D electrode **708b** and a second portion **709''** of the gate structure **709**. The location of routing tracks **706** in the second cell **730** are the same as the location of the routing tracks **706** in the FET layout **705**. Maintaining the same location of the routing tracks **706** in the first cell **720** and the second cell **730** helps to simplify routing operations, such as automatic placement and routing (APR).

(79) The first cell **720** is then processed to be formed on a first side of a substrate **710** and the second cell **730** is processed to be formed on a second side of the substrate **710** opposite to the first side. In some embodiments, the substrate **710** is similar to the substrate **210** (FIG. 2). Using the method **700**, an overall size of the amphi-FET device manufactured is reduced in comparison with manufacturing the FET layout **705** on a single side of the substrate.

(80) FIG. 7B is perspective views of a FET structure converted into an amphi-FET structure in accordance with some embodiments. The substrate **710** is not included in FIG. 7B for the sake of clarity of the drawing. The left side of FIG. 7B is an example of the FET layout **705** for manufacturing on a single side of a substrate. The horizontal cut line **715** is not part of the structure, but is included in FIG. 7B to indicate a location where the device is cut based on the method **700** (FIG. 7A). The right side of FIG. 7B is an example of an amphi-FET structure formed

by the method **700**. As seen in FIG. 7B, elements to the left of the horizontal cut line **715** form the first cell **720** on a first side of a substrate. Elements to the right of the horizontal cut line **715** form the second cell **730** on a second side of the substrate. A gate via **714**, similar to gate via **514** (FIG. 5C), electrically connects the first portion **709'** of the gate structure **709** to the second portion **709''** of the gate structure **709** through the substrate. An S/D connect via **712**, similar to the S/D connect via **512** (FIG. 5B), electrically connects the first portion **708a'** of the first S/D electrode **708a** to the second portion **708a''** of the first S/D electrode **708a** through the substrate.

(81) FIG. 7C is a perspective view of a portion of the amphi-FET structure of FIG. 7B in accordance with some embodiments. Other portions of the amphi-FET structure are omitted for the sake of clarity. FIG. 7C includes the substrate **710**. No S/D connect via extends through the substrate **710** to electrically connect to the third S/D electrode **708c**.

(82) FIG. 7D is a perspective view of a portion of the amphi-FET structure of FIG. 7B. In comparison with FIG. 7C, the first portion **709'** of the gate structure **709** and the second portion **709''** of the gate structure **709** are included. The gate via **714** extends through the substrate **710** to electrically connect the first portion **709'** of the gate structure **709** to the second portion **709''** of the gate structure.

(83) FIG. 7E is a perspective view of the amphi-FET structure of FIG. 7B including the substrate **710**. In comparison with FIG. 7D, the first portion **708a'** of the first S/D electrode **708a** and the second portion **708''** of the first S/D electrode **708a** are included. The S/D connect via **712** extends through the substrate **710** to electrically connect the first portion **708a'** of the first S/D electrode **708a** and the second portion **708''** of the first S/D electrode **708a**.

(84) FIG. 8 is view of converting neighboring cells in a layout of a FET structure **800A** to a layout of an amphi-FET structure **800B**, in accordance with some embodiments. The layout of the FET structure **800A** includes two cells having cell boundaries **802**. For clarity, only routing lines are included in the layout of the FET structure **800A**. One of ordinary skill in the art would understand that additional components, such as those described above, are part of the FET structure **800A** and the amphi-FET structure **800B**. A first conductive line **840** is in the first cell **820** and a second conductive line **850** is in the second cell **830**. A power rail **845** extends across a cell boundary **802** between the first cell **820** and the second cell **830**. The power rail **845** is called a shared power rail, in some instances.

(85) The layout of the amphi-FET structure **800B** includes the first cell **820** on a first side of a substrate **810** and the second cell **830** on the second side of the substrate **810** with cell boundaries **802'** for both cells aligned. When the layout of the FET structure **800A** is converted to a layout of an amphi-FET structure **800B**, the shared power rail is duplicated so that a first power rail **845a** is in the first cell **820** and a second power rail **845b** is in the second cell **830**. The shared power rail is duplicated instead of being cut in half in order to avoid the increase in resistance that would result from reducing the size of the power rail. While the duplication of the shared power rail means a magnitude of the size reduction for the amphi-FET structure **800B** is not as significant as if the shared power rail is cut in half, the amphi-FET structure **800B** is still smaller than the FET structure **800A**. In addition, the power consumption of the amphi-FET structure **800B** will be similar to the power consumption of the FET structure **800A** due to the similar sizes of the power rails in both structures.

(86) FIG. 9 is a view of a layout of an amphi-FET structure **900** in accordance with some embodiments. For clarity, only routing lines are included in the layout of the amphi-FET structure **900**. One of ordinary skill in the art would understand that additional components, such as those described above, are part of the amphi-FET structure **900**. The layout of the amphi-FET structure **900** includes a first cell **920** on a first side of a substrate **910** and a second cell **930** on the second side of the substrate **910** with cell boundaries **902** for both cells aligned. The layout of the amphi-FET structure **900** includes a first shared power rail **950a** in the first cell **920** and a second shared power rail **950b** in the second cell **930**. The first shared power rail **950a** and the second shared

power rail **950b** both overlap a cell boundary **902** and are accessible by an adjacent cell on the same side of the substrate as each of the corresponding share power rails.

(87) The first cell **920** further includes a first conductive line **930a** completely within the cell boundaries **902** and a first shielding line **940a** on the cell boundary **902**. The second cell **930** further includes a second conductive line **930b** completely within the cell boundaries **902** and a second shielding line **940b** on the cell boundary **902**. A shielding line is usable to reduce coupling noise between adjacent conductive lines in an interconnect structure. In a structure formed entirely on a same side of the substrate, the shielding line is not usable because using the shielding line to carry a signal would negate the shielding effect. However, when used in an amphi-FET structure, one of the first shielding line **940a** or the second shielding line **940b** is usable to carry a signal. Thus, by using the amphi-FET structure, an additional line within the interconnect structure becomes usable for routing. The additional routing options helps to reduce or avoid increases in size of the amphi-FET in order to accommodate routing of signals to the amphi-FET. In some instances, the layout of the amphi-FET **900** is called a 2.5 T horizontal cut mirror because there are five usable tracks (power rails are excluded from usable track determinations) between two cells. That is, each of the first cell **920** and the second cell **930** is considered to have 2.5 usable tracks.

(88) FIG. **10** is a view of a layout of an amphi-FET structure **1000** in accordance with some embodiments. For clarity, only routing lines are included in the layout of the amphi-FET structure **1000**. One of ordinary skill in the art would understand that additional components, such as those described above, are part of the amphi-FET structure **1000**. The layout of the amphi-FET structure **1000** includes a first cell **1020** on a first side of a substrate **1010** and a second cell **1030** on the second side of the substrate **1010** with cell boundaries **1002** for both cells aligned. The layout of the amphi-FET structure **1000** includes a first power rail **1050a** completely within the cell boundaries **1002** in the first cell **1020**. A second power rail **1050b** is completely within the cell boundaries **1002** in the second cell **1030**. The first cell **1020** further includes a first conductive line **1030a** for carrying signals into and out of the first cell **1020** and the second cell **1030** includes a second conductive line **1030b** for carrying signals into and out of the second cell **1030**. The other unlabeled conductive lines are similar to the first conductive line **1030a** and the second conductive line **1030b**. The first cell **1020** and the second cell **1030** are free of shielding lines along the cell boundaries **1002**. Thus, each of the first cell **1020** and the second cell **1030** includes four usable tracks. In some instances, the layout of the amphi-FET **1000** is called a 4 T vertical cut stack.

(89) FIG. **11** is a view of a layout of an amphi-FET structure **1100** in accordance with some embodiments. Less than all of the elements in the layout of the amphi-FET **1100** are labeled for the sake of clarity of the drawing. A legend is provided to assist with understanding of the different symbols in the layout of the amphi-FET **1100**. The layout of the amphi-FET structure **1100** includes a layout of a first cell **1120** to be formed on a first side of a substrate and a layout of a second cell **1130** to be formed on a second side of the substrate opposite the first side.

(90) The first cell **1120** includes a plurality of routing tracks **1106**. The routing tracks **1106** are locations where conductive lines are able to be formed one level farther away from the substrate than the elements in the layout of the amphi-FET structure **1100**. The first cell **1120** further includes a first active region **1115a** and a second active region **1115b** spaced from the first active region **1115a**. In some embodiments, the first active region **1115a** and the second active region **1115b** include one or more NS. The first cell **1120** further includes a first S/D electrode **1122** associated with the first active region **1115a**. The first cell **1120** further includes a first gate structure **1132a** and a second gate structure **1132b**, each of which are associated with both the first active region **1115a** and the second active region **1115b**.

(91) The second cell **1130** includes the plurality of routing tracks **1106**. The routing tracks **1106** are locations where conductive lines are able to be formed one level farther away from the substrate than the elements in the layout of the amphi-FET structure **1100**. The second cell **1130** further includes a third active region **1115a'** and a fourth active region **1115b'** spaced from the third active

region **1115a'**. In some embodiments, the third active region **1115a'** and the fourth active region **1115b'** include one or more NS. The second cell **1130** further includes a second S/D electrode **1122'** associated with the third active region **1115a'**. The second cell **1130** further includes a third gate structure **1132a'** and a fourth gate structure **1132b'**, each of which are associated with both the third active region **1115a'** and the fourth active region **1115b'**. The second cell **1130** further includes a connector **1140** electrically connecting an S/D electrode associated with the third active region **1115a'** to an S/D electrode associated with the fourth active region **1115b'**.

(92) An S/D connect via **1112** electrically connects the first S/D electrode **1122** to the second S/D electrode **1122'** through the substrate. A gate via **1114** electrically connects the first gate structure **1132a** to the third gate structure **1132a'** through the substrate. One of ordinary skill in the art would understand that the components of the layout of the amphi-FET **1100** are formed using materials and processes described above, which are not repeated here for the sake of brevity.

(93) FIG. **12** is a view of a layout of an amphi-FET structure **1200** in accordance with some embodiments. Less than all of the elements in the layout of the amphi-FET **1200** are labeled for the sake of clarity of the drawing. A legend is provided to assist with understanding of the different symbols in the layout of the amphi-FET **1200**. In comparison with the layout of amphi-FET **1100** (FIG. **11**), a size of the active regions in the layout of the amphi-FET **1200** are smaller. The smaller active regions reduces a size of the layout of the amphi-FET **1200**, but also reduces switching speed. The layout of the amphi-FET structure **1200** includes a layout of a first cell **1220** to be formed on a first side of a substrate; and a layout of a second cell **1230** to be formed on a second side of the substrate opposite the first side.

(94) The first cell **1220** includes a plurality of routing tracks **1206**. The routing tracks **1206** are locations where conductive lines are able to be formed one level farther away from the substrate than the elements in the layout of the amphi-FET structure **1200**. The first cell **1220** further includes a first active region **1215a** and a second active region **1215b** spaced from the first active region **1215a**. In some embodiments, the first active region **1215a** and the second active region **1215b** include one or more NS. The first cell **1220** further includes a first S/D electrode **1222** associated with the first active region **1215a**. The first cell **1220** further includes a first gate structure **1232a** and a second gate structure **1232b**, each of which are associated with both the first active region **1215a** and the second active region **1215b**.

(95) The second cell **1230** includes the plurality of routing tracks **1206**. The routing tracks **1206** are locations where conductive lines are able to be formed one level farther away from the substrate than the elements in the layout of the amphi-FET structure **1200**. The second cell **1230** further includes a third active region **1215a'** and a fourth active region **1215b'** spaced from the third active region **1215a'**. In some embodiments, the third active region **1215a'** and the fourth active region **1215b'** include one or more NS. The second cell **1230** further includes a second S/D electrode **1222'** associated with the third active region **1215a'**. The second cell **1230** further includes a third gate structure **1232a'** and a fourth gate structure **1232b'**, each of which are associated with both the third active region **1215a'** and the fourth active region **1215b'**. The second cell **1230** further includes a connector **1240** electrically connecting an S/D electrode associated with the third active region **1215a'** to an S/D electrode associated with the fourth active region **1215b'**.

(96) An S/D connect via **1212** electrically connects the first S/D electrode **1222** to the second S/D electrode **1222'** through the substrate. A gate via **1214** electrically connects the first gate structure **1232a** to the third gate structure **1232a'** through the substrate. One of ordinary skill in the art would understand that the components of the layout of the amphi-FET **1200** are formed using materials and processes described above, which are not repeated here for the sake of brevity.

(97) FIG. **13** is a view of a layout of an amphi-FET structure **1300** in accordance with some embodiments. Less than all of the elements in the layout of the amphi-FET **1300** are labeled for the sake of clarity of the drawing. A legend is provided to assist with understanding of the different symbols in the layout of the amphi-FET **1300**. The layout of the amphi-FET structure **1300**

includes a layout of a first cell **1320** to be formed on a first side of a substrate; and a layout of a second cell **1330** to be formed on a second side of the substrate opposite to the first side.

(98) The first cell **1320** includes a plurality of routing tracks **1306**. The routing tracks **1306** are locations where conductive lines are able to be formed one level farther away from the substrate than the elements in the layout of the amphi-FET structure **1300**. The first cell **1320** further includes a first active region **1315a** and a second active region **1315b** spaced from the first active region **1315a**. In some embodiments, the first active region **1315a** and the second active region **1315b** include one or more NS. The first cell **1320** further includes a first S/D electrode **1322** associated with the first active region **1315a**. The first cell **1320** further includes a first gate structure **1332a** and a second gate structure **1332b**, each of which are associated with both the first active region **1315a** and the second active region **1315b**.

(99) The second cell **1330** includes the plurality of routing tracks **1306**. The routing tracks **1306** are locations where conductive lines are able to be formed one level farther away from the substrate than the elements in the layout of the amphi-FET structure **1300**. The second cell **1330** further includes a third active region **1315a'** and a fourth active region **1315b'** spaced from the third active region **1315a'**. In some embodiments, the third active region **1315a'** and the fourth active region **1315b'** include one or more NS. The second cell **1330** further includes a second S/D electrode **1322'** associated with the third active region **1315a'**. The second cell **1330** further includes a third gate structure **1332a'** and a fourth gate structure **1332b'**, each of which are associated with both the third active region **1315a'** and the fourth active region **1315b'**. The second cell **1330** further includes a connector **1340** electrically connecting an S/D electrode associated with the third active region **1315a'** to an S/D electrode associated with the fourth active region **1315b'**.

(100) An S/D connect via **1312** electrically connects the first S/D electrode **1322** to the second S/D electrode **1322'** through the substrate. A gate via **1314** electrically connects the first gate structure **1332a** to the third gate structure **1332a'** through the substrate. One of ordinary skill in the art would understand that the components of the layout of the amphi-FET **1300** are formed using materials and processes described above, which are not repeated here for the sake of brevity.

(101) FIG. **14** is a block diagram of an integrated circuit (IC) manufacturing system **1400**, and an IC manufacturing flow associated therewith, in accordance with some embodiments. In some embodiments, based on a layout diagram, at least one of (A) one or more semiconductor masks or (B) at least one component in a layer of a semiconductor integrated circuit is fabricated using IC manufacturing system **1400**.

(102) In FIG. **14**, IC manufacturing system **1400** includes entities, such as a design house **1420**, a mask house **1430**, and an IC manufacturer/fabricator ("fab") **1450**, that interact with one another in the design, development, and manufacturing cycles and/or services related to manufacturing an IC device **1460**. The entities in IC manufacturing system **1400** are connected by a communications network. In some embodiments, the communications network is a single network. In some embodiments, the communications network is a variety of different networks, such as an intranet and the Internet. The communications network includes wired and/or wireless communication channels. Each entity interacts with one or more of the other entities and provides services to and/or receives services from one or more of the other entities. In some embodiments, two or more of design house **1420**, mask house **1430**, and IC fab **1450** is owned by a single larger company. In some embodiments, two or more of design house **1420**, mask house **1430**, and IC fab **1450** coexist in a common facility and use common resources.

(103) Design house (or a design team) **1420** generates an IC design layout diagram **1422**. IC design layout diagram **1422** includes various geometrical patterns designed for an IC device **1460**. The geometrical patterns correspond to patterns of metal, oxide, or semiconductor layers that make up the various components of IC device **1460** to be fabricated. The various layers combine to form various IC features. For example, a portion of IC design layout diagram **1422** includes various IC features, such as an active region, gate electrode, source and drain region of the channel bar, source

electrode, and drain electrode, metal lines or vias of an interlayer interconnection, and openings for bonding pads, to be formed in a semiconductor substrate (such as a silicon wafer) and various material layers disposed on the semiconductor substrate. Design house **1420** implements a proper design procedure to form IC design layout diagram **1422**. The design procedure includes one or more of logic design, physical design or place and route. IC design layout diagram **1422** is presented in one or more data files having information of the geometrical patterns. For example, IC design layout diagram **1422** can be expressed in a GDSII file format or DFII file format.

(104) Mask house **1430** includes mask data preparation **1432** and mask fabrication **1444**. Mask house **1430** uses IC design layout diagram **1422** to manufacture one or more masks **1445** to be used for fabricating the various layers of IC device **1460** according to IC design layout diagram **1422**. Mask house **1430** performs mask data preparation **1432**, where IC design layout diagram **1422** is translated into a representative data file (RDF). Mask data preparation **1432** provides the RDF to mask fabrication **1444**. Mask fabrication **1444** includes a mask writer. A mask writer converts the RDF to an image on a substrate, such as a mask **1445** or a semiconductor wafer **1453**. The design layout diagram **1422** is manipulated by mask data preparation **1432** to comply with particular characteristics of the mask writer and/or requirements of IC fab **1450**. In FIG. **14**, mask data preparation **1432** and mask fabrication **1444** are illustrated as separate elements. In some embodiments, mask data preparation **1432** and mask fabrication **1444** can be collectively referred to as mask data preparation.

(105) In some embodiments, mask data preparation **1432** includes optical proximity correction (OPC) which uses lithography enhancement techniques to compensate for image errors, such as those that can arise from diffraction, interference, other process effects and the like. OPC adjusts IC design layout diagram **1422**. In some embodiments, mask data preparation **1432** includes further resolution enhancement techniques (RET), such as off-axis illumination, sub-resolution assist features, phase-shifting masks, other suitable techniques, and the like or combinations thereof. In some embodiments, inverse lithography technology (ILT) is also used, which treats OPC as an inverse imaging problem.

(106) In some embodiments, mask data preparation **1432** includes a mask rule checker (MRC) that checks the IC design layout diagram **1422** that has undergone processes in OPC with a set of mask creation rules which contain certain geometric and/or connectivity restrictions to ensure sufficient margins, to account for variability in semiconductor manufacturing processes, and the like. In some embodiments, the MRC modifies the IC design layout diagram **1422** to compensate for limitations during mask fabrication **1444**, which may undo part of the modifications performed by OPC in order to meet mask creation rules.

(107) In some embodiments, mask data preparation **1432** includes lithography process checking (LPC) that simulates processing that will be implemented by IC fab **1450** to fabricate IC device **1460**. LPC simulates this processing based on IC design layout diagram **1422** to create a simulated manufactured device, such as IC device **1460**. The processing parameters in LPC simulation can include parameters associated with various processes of the IC manufacturing cycle, parameters associated with tools used for manufacturing the IC, and/or other aspects of the manufacturing process. LPC takes into account various factors, such as aerial image contrast, depth of focus (DOF), mask error enhancement factor (MEEF), other suitable factors, and the like or combinations thereof. In some embodiments, after a simulated manufactured device has been created by LPC, if the simulated device is not close enough in shape to satisfy design rules, OPC and/or MRC are repeated to further refine IC design layout diagram **1422**.

(108) It should be understood that the above description of mask data preparation **1432** has been simplified for the purposes of clarity. In some embodiments, mask data preparation **1432** includes additional features such as a logic operation (LOP) to modify the IC design layout diagram **1422** according to manufacturing rules. Additionally, the processes applied to IC design layout diagram **1422** during mask data preparation **1432** may be executed in a variety of different orders.

(109) After mask data preparation **1432** and during mask fabrication **1444**, a mask **1445** or a group of masks **1445** are fabricated based on the modified IC design layout diagram **1422**. In some embodiments, mask fabrication **1444** includes performing one or more lithographic exposures based on IC design layout diagram **1422**. In some embodiments, an electron-beam (e-beam) or a mechanism of multiple e-beams is used to form a pattern on a mask (photomask or reticle) **1445** based on the modified IC design layout diagram **1422**. Mask **1445** can be formed in various technologies. In some embodiments, mask **1445** is formed using binary technology. In some embodiments, a mask pattern includes opaque regions and transparent regions. A radiation beam, such as an ultraviolet (UV) beam, used to expose the image sensitive material layer (e.g., photoresist) which has been coated on a wafer, is blocked by the opaque region and transmits through the transparent regions. In one example, a binary mask version of mask **1445** includes a transparent substrate (e.g., fused quartz) and an opaque material (e.g., chromium) coated in the opaque regions of the binary mask. In another example, mask **1445** is formed using a phase shift technology. In a phase shift mask (PSM) version of mask **1445**, various features in the pattern formed on the phase shift mask are configured to have proper phase difference to enhance the resolution and imaging quality. In various examples, the phase shift mask can be attenuated PSM or alternating PSM. The mask(s) generated by mask fabrication **1444** is used in a variety of processes. For example, such a mask(s) is used in an ion implantation process to form various doped regions in semiconductor wafer **1453**, in an etching process to form various etching regions in semiconductor wafer **1453**, and/or in other suitable processes.

(110) IC fab **1450** is an IC fabrication business that includes one or more manufacturing facilities for the fabrication of a variety of different IC products. In some embodiments, IC Fab **1450** is a semiconductor foundry. For example, there may be a manufacturing facility for the front end fabrication of a plurality of IC products (front-end-of-line (FEOL) fabrication), while a second manufacturing facility may provide the back end fabrication for the interconnection and packaging of the IC products (back-end-of-line (BEOL) fabrication), and a third manufacturing facility may provide other services for the foundry business.

(111) IC fab **1450** includes fabrication tools **1452** configured to execute various manufacturing operations on semiconductor wafer **1453** such that IC device **1460** is fabricated in accordance with the mask(s), e.g., mask **1445**. In various embodiments, fabrication tools **1452** include one or more of a wafer stepper, an ion implanter, a photoresist coater, a process chamber, e.g., a CVD chamber or LPCVD furnace, a CMP system, a plasma etch system, a wafer cleaning system, or other manufacturing equipment capable of performing one or more suitable manufacturing processes as discussed herein.

(112) IC fab **1450** uses mask(s) **1445** fabricated by mask house **1430** to fabricate IC device **1460**. Thus, IC fab **1450** at least indirectly uses IC design layout diagram **1422** to fabricate IC device **1460**. In some embodiments, semiconductor wafer **1453** is fabricated by IC fab **1450** using mask(s) **1445** to form IC device **1460**. In some embodiments, the IC fabrication includes performing one or more lithographic exposures based at least indirectly on IC design layout diagram **1422**. Semiconductor wafer **1453** includes a silicon substrate or other proper substrate having material layers formed thereon. Semiconductor wafer **1453** further includes one or more of various doped regions, dielectric features, multilevel interconnects, and the like (formed at subsequent manufacturing steps).

(113) FIG. **15** is a block diagram of an electronic design automation (EDA) system **1500** in accordance with some embodiments.

(114) In some embodiments, EDA system **1500** includes an APR system. Methods described herein of designing layout diagrams represent wire routing arrangements, in accordance with one or more embodiments, are implementable, for example, using EDA system **1500**, in accordance with some embodiments.

(115) In some embodiments, EDA system **1500** is a general purpose computing device including a

processor **1502** and a non-transitory, computer-readable storage medium **1504**. Computer-readable storage medium **1504** is, amongst other things, encoded with, i.e., stores, computer program code **1506**, i.e., a set of executable instructions. Execution of computer program code **1506** by processor **1502** represents (at least in part) an EDA tool which implements a portion or all of the methods described herein in accordance with one or more embodiments (hereinafter, the noted processes and/or methods).

(116) Processor **1502** is electrically coupled to computer-readable storage medium **1504** via a bus **1508**. Processor **1502** is also electrically coupled to an I/O interface **1510** by bus **1508**. A network interface **1512** is also electrically connected to processor **1502** via bus **1508**. Network interface **1512** is connected to a network **1514**, so that processor **1502** and computer-readable storage medium **1504** are capable of connecting to external elements via network **1514**. Processor **1502** is configured to execute computer program code **1506** encoded in computer-readable storage medium **1504** in order to cause EDA system **1500** to be usable for performing a portion or all of the noted processes and/or methods. In one or more embodiments, processor **1502** is a central processing unit (CPU), a multi-processor, a distributed processing system, an application specific integrated circuit (ASIC), and/or a suitable processing unit.

(117) In one or more embodiments, computer-readable storage medium **1504** is an electronic, magnetic, optical, electromagnetic, infrared, and/or a semiconductor system (or apparatus or device). For example, computer-readable storage medium **1504** includes a semiconductor or solid-state memory, a magnetic tape, a removable computer diskette, a random access memory (RAM), a read-only memory (ROM), a rigid magnetic disk, and/or an optical disk. In one or more embodiments using optical disks, computer-readable storage medium **1504** includes a compact disk-read only memory (CD-ROM), a compact disk-read/write (CD-R/W), and/or a digital video disc (DVD).

(118) In one or more embodiments, storage medium **1504** stores computer program code **1506** configured to cause EDA system **1500** (where such execution represents (at least in part) the EDA tool) to be usable for performing a portion or all of the noted processes and/or methods. In one or more embodiments, storage medium **1504** also stores information which facilitates performing a portion or all of the noted processes and/or methods. In one or more embodiments, storage medium **1504** stores library **1507** of standard cells including such standard cells as disclosed herein.

(119) EDA system **1500** includes I/O interface **1510**. I/O interface **1510** is coupled to external circuitry. In one or more embodiments, I/O interface **1510** includes a keyboard, keypad, mouse, trackball, trackpad, touchscreen, and/or cursor direction keys for communicating information and commands to processor **1502**.

(120) EDA system **1500** also includes network interface **1512** coupled to processor **1502**. Network interface **1512** allows EDA system **1500** to communicate with network **1514**, to which one or more other computer systems are connected. Network interface **1512** includes wireless network interfaces such as BLUETOOTH, WIFI, WIMAX, GPRS, or WCDMA; or wired network interfaces such as ETHERNET, USB, or IEEE-1364. In one or more embodiments, a portion or all of noted processes and/or methods, is implemented in two or more EDA systems **1500**.

(121) EDA system **1500** is configured to receive information through I/O interface **1510**. The information received through I/O interface **1510** includes one or more of instructions, data, design rules, libraries of standard cells, and/or other parameters for processing by processor **1502**. The information is transferred to processor **1502** via bus **1508**. EDA system **1500** is configured to receive information related to a UI through I/O interface **1510**. The information is stored in storage medium **1504** as user interface (UI) **1542**.

(122) In some embodiments, a portion or all of the noted processes and/or methods is implemented as a standalone software application for execution by a processor. In some embodiments, a portion or all of the noted processes and/or methods is implemented as a software application that is a part of an additional software application. In some embodiments, a portion or all of the noted processes

and/or methods is implemented as a plug-in to a software application. In some embodiments, at least one of the noted processes and/or methods is implemented as a software application that is a portion of an EDA tool. In some embodiments, a portion or all of the noted processes and/or methods is implemented as a software application that is used by EDA system **1500**. In some embodiments, a layout diagram which includes standard cells is generated using a tool such as VIRTUOSO® available from CADENCE DESIGN SYSTEMS, Inc., or another suitable layout generating tool.

(123) In some embodiments, the processes are realized as functions of a program stored in a non-transitory computer readable recording medium. Examples of a non-transitory computer readable recording medium include, but are not limited to, external/removable and/or internal/built-in storage or memory unit, e.g., one or more of an optical disk, such as a DVD, a magnetic disk, such as a hard disk, a semiconductor memory, such as a ROM, a RAM, a memory card, and the like.

(124) Details regarding an integrated circuit (IC) manufacturing system (e.g., EDA system **1500** of FIG. **15**), and an IC manufacturing flow associated therewith are found, e.g., in U.S. Pat. No. 9,256,709, granted Feb. 9, 2016, U.S. Pre-Grant Publication No. 20178429, published Oct. 1, 2015, U.S. Pre-Grant Publication No. 20140040838, published Feb. 6, 2014, and U.S. Pat. No. 7,260,442, granted Aug. 21, 2007, the entireties of each of which are hereby incorporated by reference.

(125) An aspect of this description relates to a semiconductor device. The semiconductor device includes a substrate. The semiconductor device further includes a first gate structure on a first side of the substrate. The semiconductor device further includes a second gate structure on a second side of the substrate, wherein the first side is opposite the second side. The semiconductor device further includes a gate via extending through the substrate, wherein the gate via directly connects to the first gate structure, and the gate via directly connects to the second gate structure. In some embodiments, the first gate structure is configured to control conductivity of a portion of a first active region, and the second gate structure is configured to control conductivity of a portion of a second active region. In some embodiments, the gate via is offset from the first active region in a direction parallel to a top surface of the substrate in a cross-sectional view. In some embodiments, the semiconductor device further includes a first source/drain (S/D) electrode electrically connected to the first active region; and a second S/D electrode electrically connected to the second active region. In some embodiments, the semiconductor device further includes a S/D via electrically connecting the first S/D electrode to the second S/D electrode, wherein the S/D via extends through the substrate. In some embodiments, a width of the S/D via measured in a first direction parallel to a top surface of the substrate is equal to a width of the second S/D electrode. In some embodiments, the first S/D electrode extends beyond the second S/D electrode in the first direction. In some embodiments, the width of the S/D via is equal to a width of the first S/D electrode. In some embodiments, the semiconductor device further includes a first power rail on the first side of the substrate, wherein the first power rail is electrically connected to the first gate structure. In some embodiments, the semiconductor device further includes a second power rail on the second side of the substrate, wherein the second power rail is electrically connected to the second gate structure.

(126) An aspect of this description relates to a method of making a semiconductor device. The method includes forming a first source/drain (S/D) electrode on a first side of a substrate. The method further includes forming an S/D connect via extending through the substrate. The method further includes forming a second S/D electrode on a second side of the substrate, wherein the first side is opposite the second side, and the S/D connect directly contacts both the first S/D electrode and the second S/D electrode. In some embodiments, the method further includes forming a first gate structure on the first side of the substrate, wherein the first gate structure is configured to control conductivity of a first active region electrically connected to the first S/D electrode. In some embodiments, the method further includes forming a second gate structure on the second side of the substrate, wherein the second gate structure is configured to control conductivity of a second active region electrically connected to the second S/D electrode. In some embodiments, the method

further includes a gate via extending through the substrate, wherein the gate via electrically connects the first gate structure to the second gate structure. In some embodiments, forming the gate via comprises forming the gate via in direct contact with the first gate structure and in direct contact with the second gate structure. In some embodiments, forming the second S/D electrode comprises forming the second S/D electrode having a same width as the S/D connect via. In some embodiments, forming the second S/D electrode comprises forming the second S/D electrode having a different width from the first S/D electrode.

(127) An aspect of this description relates to a semiconductor device. The semiconductor device includes a substrate. The semiconductor device further includes a first active region on a first side of the substrate. The semiconductor device further includes a second active region on a second side of the substrate opposite the first side. The semiconductor device further includes a first electrode surrounding a portion of the first active region. The semiconductor device further includes a second electrode surrounding a portion of the second active region. The semiconductor device further includes a via extending through the substrate, wherein the via directly connects to the first electrode, and the via directly connects to the second electrode. In some embodiments, the first electrode is a gate electrode. In some embodiments, the first electrode is a source/drain (S/D) electrode.

(128) The foregoing outlines features of several embodiments so that those skilled in the art may better understand the aspects of the present disclosure. Those skilled in the art should appreciate that they may readily use the present disclosure as a basis for designing or modifying other processes and structures for carrying out the same purposes and/or achieving the same advantages of the embodiments introduced herein. Those skilled in the art should also realize that such equivalent constructions do not depart from the spirit and scope of the present disclosure, and that they may make various changes, substitutions, and alterations herein without departing from the spirit and scope of the present disclosure.

Claims

1. A semiconductor device comprising: a substrate; a first gate structure on a first side of the substrate; a second gate structure on a second side of the substrate, wherein the first side is opposite the second side; and a gate via extending through the substrate, wherein the gate via directly connects to the first gate structure, and the gate via directly connects to the second gate structure.
2. The semiconductor device of claim 1, wherein the first gate structure is configured to control conductivity of a portion of a first active region, and the second gate structure is configured to control conductivity of a portion of a second active region.
3. The semiconductor device of claim 2, wherein the gate via is offset from the first active region in a direction parallel to a top surface of the substrate in a cross-sectional view.
4. The semiconductor device of claim 2, further comprising: a first source/drain (S/D) electrode electrically connected to the first active region; and a second S/D electrode electrically connected to the second active region.
5. The semiconductor device of claim 4, further comprising a S/D via electrically connecting the first S/D electrode to the second S/D electrode, wherein the S/D via extends through the substrate.
6. The semiconductor device of claim 5, wherein a width of the S/D via measured in a first direction parallel to a top surface of the substrate is equal to a width of the second S/D electrode.
7. The semiconductor device of claim 6, wherein the first S/D electrode extends beyond the second S/D electrode in the first direction.
8. The semiconductor device of claim 6, wherein the width of the S/D via is equal to a width of the first S/D electrode.
9. The semiconductor device of claim 1, further comprising a first power rail on the first side of the substrate, wherein the first power rail is electrically connected to the first gate structure.

10. The semiconductor device of claim 9, further comprising a second power rail on the second side of the substrate, wherein the second power rail is electrically connected to the second gate structure.

11. A method of making a semiconductor device, the method comprising: forming a first source/drain (S/D) electrode on a first side of a substrate; forming an S/D connect via extending through the substrate; and forming a second S/D electrode on a second side of the substrate, wherein the first side is opposite the second side, and the S/D connect directly contacts both the first S/D electrode and the second S/D electrode.

12. The method of claim 11, further comprising forming a first gate structure on the first side of the substrate, wherein the first gate structure is configured to control conductivity of a first active region electrically connected to the first S/D electrode.

13. The method of claim 12, further comprising forming a second gate structure on the second side of the substrate, wherein the second gate structure is configured to control conductivity of a second active region electrically connected to the second S/D electrode.

14. The method of claim 13, further comprising forming a gate via extending through the substrate, wherein the gate via electrically connects the first gate structure to the second gate structure.

15. The method of claim 14, wherein forming the gate via comprises forming the gate via in direct contact with the first gate structure and in direct contact with the second gate structure.

16. The method of claim 11, wherein forming the second S/D electrode comprises forming the second S/D electrode having a same width as the S/D connect via.

17. The method of claim 11, wherein forming the second S/D electrode comprises forming the second S/D electrode having a different width from the first S/D electrode.

18. A semiconductor device comprising: a substrate; a first active region on a first side of the substrate; a second active region on a second side of the substrate opposite the first side; a first electrode surrounding a portion of the first active region; a second electrode surrounding a portion of the second active region; and a via extending through the substrate, wherein the via directly connects to the first electrode, and the via directly connects to the second electrode.

19. The semiconductor device of claim 18, wherein the first electrode is a gate electrode.

20. The semiconductor device of claim 18, wherein the first electrode is a source/drain (S/D) electrode.
